# TRACE-RPG: Auditable Neuro-Symbolic Commit Gates for Interactive Game Worlds

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***Abstract*—TRACE-RPG treats generated game events as untrusted proposals. Strict parsing, external policy, seven predicates in six families, bounded repair, and pre-commit revalidation gate mutation; typed-error $\rho$ reads candidates/errors, not authoritative state. Unkeyed SHA-256 links, semantic replay, and complete accounting audit outcomes. Separate evidence covers one-world offline mechanism conformance, single-model live screening, non-authoritative closed-world graph simulation, and participant-free Godot engineering. At $K{=}1$, guided/blind commits differed 5/5 versus 0/5 only in a constructed repairable policy-blind regime. No lane establishes population efficacy, model superiority, player experience, or commercial-engine performance.**

***Index Terms*—game AI, interactive narrative, symbolic validation, constrained generation, reproducibility**

## I. Introduction

GENERATED events may require absent items, assert unknown facts, disclose secrets, introduce unauthorized effects, or regress quests. Plausibility [1], [2] and quest/entity grounding [3], [4] confer no authority: **only TRACE-RPG's external deterministic gate authorizes mutation** (Fig. 1).

**C1—Contracts:** versioned parsing; **C2—Controller:** validation, bounded repair, unchanged fallback, revalidation; **C3—Audit:** linked replay/continuity; **C4—Harness:** complete accounting without mandatory proposal traces; **C5—Repair:** typed $\rho(a, E)$ versus blind retry/state-reading. Tables I–V: C1–C3/E1/ENG1, C4/E1, C5/E1/E2. NPC-ENG1/NPC-SCR1 are descriptive, not C6/C-RESULT or core Python–hard-policy/normal-play evidence.

## II. Related Work and Research Gap

Grounding/believability spans quest/entity dialogue [3], graph quests [4], memory–reflection–planning [5], trainable roles [6]. Player/dialogue studies [7], [8], [9] and role-sensitive stability–improvisation (2025 preprint) [10] separate admissibility from experience.

Engagement/generation motivates unimplemented, non-authoritative future affect [11], [12], [13].

State/embodied/agent-game-content environments [1], [2], [14], [15] and heterogeneous/process/social/long-horizon benchmarks [16], [17], [18], [19] complement human playtesting [20]. NCP-Bench measures narrative commitment/fact consistency, not our player outcomes [21]. No reviewed protocol combined authorization, typed repair, replay, accounting; this is not a first-ever claim.

Structure [22], [23], [24] differs from authorization. Sound critics (LLM-Modulo) [25], affordances (SayCan) [26], trigger–predicate rules (AgentSpec) [27], tool preconditions (Setting the DC) [28], generated preconditions/effects (STORY2GAME) [29], LLM rule refinement (PANGeA) [30], incremental-world validation (IVIE) [31], authored transformations [32], and JSONSchemaBench structural compliance/coverage/quality [33] do not resolve our transaction/audit/repair combination or semantic authorization/superiority beyond encoded predicates.

Subgraph retrieval/long-term memory [34], [35], [36] has no commit authority; benefits are unmeasured.

Process, sparse-run, reproducibility, environmental reporting motivate frozen artifacts/failure retention [17], [37], [38], [39]. Human/LLM evaluation needs construct/rater/bias controls, justified margins/random effects [40], [41], [42], [43], [44], [45]. We replay admitted transitions, not experience reuse [46]; leakage checks protect denominators, not efficacy [47].

C1–C5 combine existing enforcement [27], game-state management [28], [29], feedback repair [48], [49], [50], [51], replay [46], action interposition [52], preconditions/effects [53], transition enforcement [54], abort [55], social logic [56], belief-aware planning [57]; none is claimed invented. Reporting omissions do not prove implementation absence.

TABLE I
Contributions, Prior Topics, Evidence Lanes, and Inference Ceilings (Generated from the Structured Matrices)

| ID | Contribution | Prior topics (refs) | Lanes | Inference ceiling |
|---|---|---|---|---|
| C1 | Trust-boundary contracts | T1, T2, T3 (17) | E1, ENG1 | encoded-field contract conformance |
| C2 | Validate–repair–commit controller | T1, T8 (14) | E1, ENG1 | mechanism behavior on frozen cases |
| C3 | Audit-linked evidence layer | T4, T7, T8 (6) | E1, ENG1 | integrity/replay on named fixtures |
| C4 | Assignment-complete harness | T1, T4, T6, T7 (20) | E1 | exact designed-case accounting |
| C5 | Counterexample-guided operator $\rho$ | T1, T8, T9 (8) | E1, E2 | frozen classes + one regime screen |

T1 = grounded game worlds; T2 = retrieval/memory agents; T3 = structured generation; T4 = environments/benchmarks; T5 = player experience; T6 = human/LLM evaluation; T7 = statistics/reproducibility; T8 = planning/runtime interposition; T9 = feedback repair. Lanes are defined in Table V. All rows are authored-fixture verified; none supports an efficacy claim.

## III. Problem Definition and Guarantee Boundary

$c_t = (G_t, q_t)$ stores typed snapshot $G_t$ and external action-policy/disclosure/quest-stage constraints $q_t$. Predicates neither read $h_{<t}$ nor reconstruct `WorldState` from logs. Transitions update $G_t$, not $q_t$; policy rotation is excluded. Retrieval, scores, rationales, memory, future affect have no authority.

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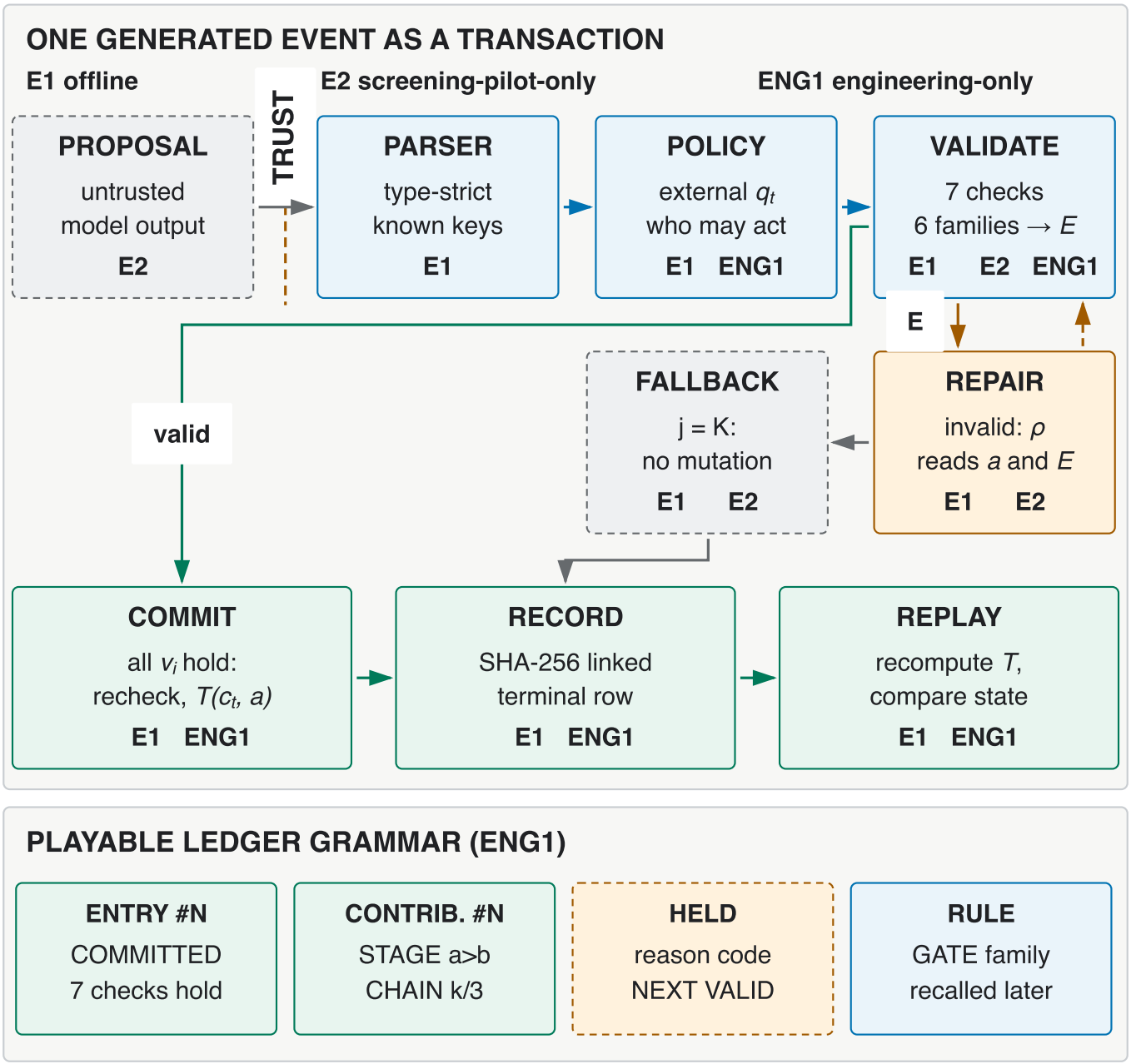



Fig. 1. Policy/$c_t$ gate proposals; errors feed bounded $\rho$ (dashed) or unchanged fallback. Passing candidates are rechecked, committed, recorded, replayed. E1/E2/legacy ENG1 tags exclude additive NPC `npc-goal-recovery/1.1.0`: Godot-authoritative intent/planning, not core Python–hard-policy round trip. Below: committed-snapshot ledger grammar.

Strict parsing of untrusted candidate $a_t$ enforces the shared key/schema contract, rejecting unknown top-level fields. Validation returns

$$V(c_t, a_t) = (\boldsymbol{v}, E_t), \quad \boldsymbol{v} = (v_{\text{policy}}, v_{\text{pre}}, v_{\text{reach}}, v_{\text{know}}, v_{\text{disc}}, v_{\text{elig}}, v_{\text{stage}}) \in \{0,1\}^7, \tag{1}$$

where $E_t$ is a structured error set. Mutation is

$$c_{t+1} = \begin{cases} T(c_t, a_t), & \bigwedge_{i=1}^{7}(v_i = 1), \\ c_t, & \text{otherwise.} \end{cases} \tag{2}$$

Fixtures test implementation invariants, not theorems. **I1:** validation exhaustion or pre-commit proposal/controller failure preserves state. **I2:** returning callbacks and $K \geq 0$ bound recorded attempts to $K + 1$, plus pre-commit revalidation; without enforced deadlines, wall-clock termination is not guaranteed. **I3:** accepted `CandidateAction.effects` stay within `ActionPolicy.allowed_effects`, including authorized quest-stage targets. **I4:** frozen schemas/deterministic transitions enable terminal-state reconstruction.

Assumptions: correct policy, faithful extraction, deterministic versions, returning callbacks, single-process file ownership.

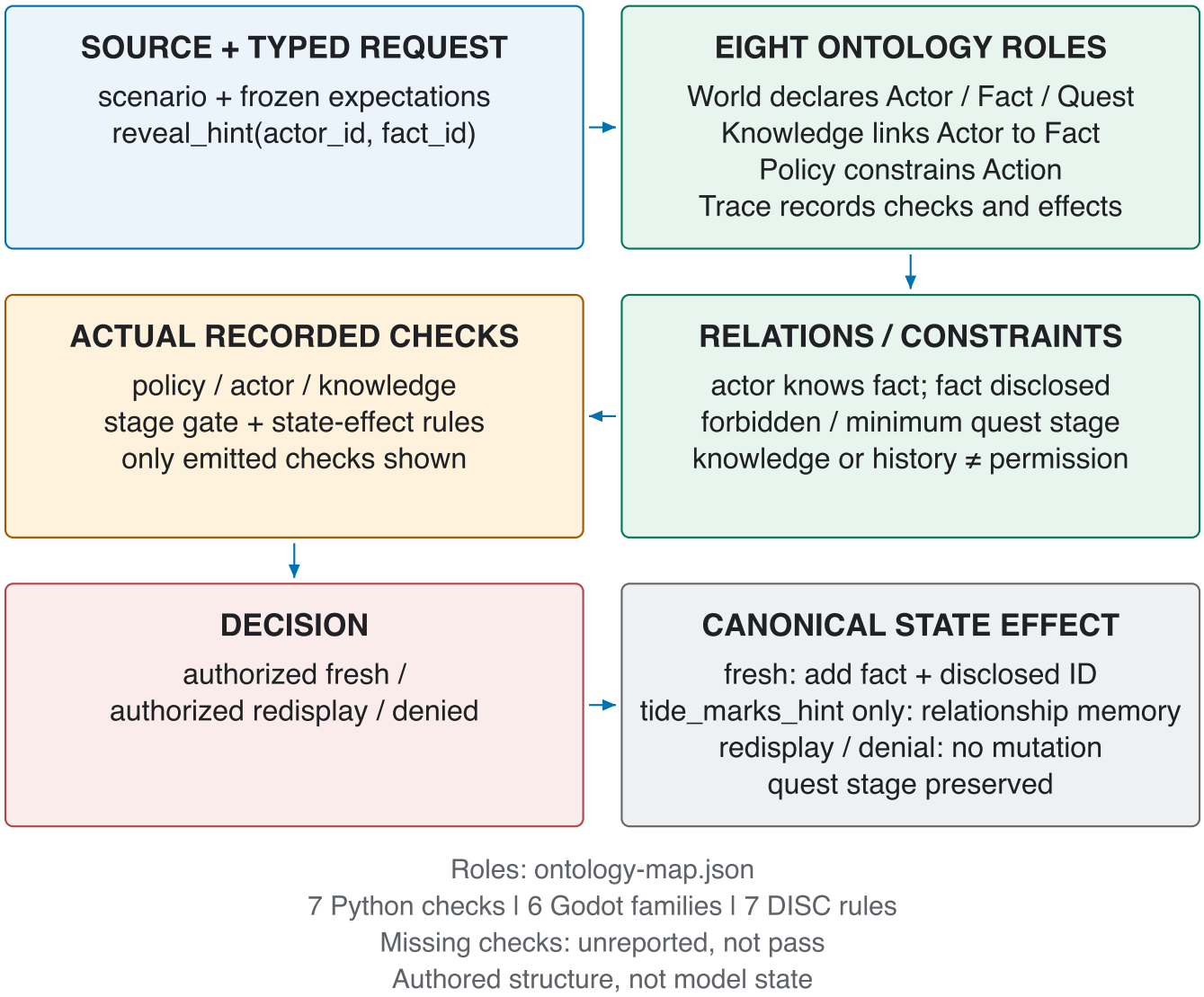



Fig. 2. Eight-role ontology structure (Table III): World, Actor, Knowledge, Fact, Policy, Quest, Action, Trace connect declared source fields to authored checks and observed decisions. Only authorized transitions change canonical state; the diagram is not live capture or hidden-state evidence.

Replay checks structured transitions, not repair provenance, prompts, stochastic generation, or undeclared natural-language implications.

## IV. TRACE-RPG Architecture and Algorithms

### A. Trust Boundary and Contracts

External models/adapters propose; parsing rejects ambiguous types/unknown keys; independent policy defines preconditions/effects. The versioned bridge carries observations, proposals, validations, rejections, fallbacks, commits (Fig. 1). Unknown-key parity establishes neither full parser equivalence nor commercial portability. Python has seven deterministic checks; Godot has six state-relative families (eligibility/stage combined), separate from seven DISC rules.

### B. Fixed-goal NPC integration

Under `npc-goal-recovery/1.1.0`, Codex proposes intent; Python checks public claims/plans; Godot alone writes state. The fixed harbor-restoration/authorized-tide-lead goal is not model-editable. Write bindings cover session, epoch, request, scenario, NPC, goal, version, policy/config hashes, revision/state hash, deadline/budget, and issued action. Service: loopback `127.0.0.1:8768`; browser origins: `localhost`,

TABLE II
Encoded Validator Boundary

| Predicate | Rejects | On failure |
|---|---|---|
| Action policy | unknown action/type | unchanged |
| Precondition | absent/false requirement | unchanged |
| Reachability | inaccessible object | unchanged |
| NPC knowledge | undeclared known fact | unchanged |
| Disclosure | forbidden fact | unchanged |
| Quest eligibility | ineligible quest action | unchanged |
| Quest stage | regressive or unauthorized stage | unchanged |

TABLE III
EIGHT ONTOLOGY ROLES AND SOURCE RELATIONSHIPS: READ-ONLY `game-track/web/dashboard/ontology-map.json` BINDS DECLARATIONS/FROZEN EXPECTATIONS, NOT INFERRED LIVE STATE OR AUTHORITY

| Role | Source field or relationship example |
|---|---|
| World | `initial_state.world.object_locations`: `signal_lens`→`lamp_store`. |
| Actor | `initial_state.npcs.captain_mira`; sole `scope.eligible_actor_ids` member. |
| Knowledge | Actor `knowledge`→canonical Fact IDs, not permission. |
| Fact | `initial_state.facts`: `dock_saved`, `lighthouse_dark`; `keeper_betrayal` is an ID, not secret payload. |
| Policy | `disclosure_policy`: forbids `keeper_betrayal`; `tide_marks_hint` needs minimum quest stage 2. |
| Quest | `initial_state.quest`: stage 0, flag `dock_saved`; disclosure preserves stage. |
| Action | `reveal_hint(actor_id, fact_id)` adds Fact/disclosed ID; only tide hint adds relationship memory. Fresh revision $+1$; redisplay unchanged. |
| Trace | `prior_state_hash`, `next_state_hash`, `rule_ids`, `template_id`, `case_id` bind packet/version to evidence; absent fields cannot establish PASS. |

1: $a \leftarrow \text{Propose}(c)$; $j \leftarrow 0$
2: **repeat**
3: $(\text{ok}, E) \leftarrow V(c, a)$; $\text{Record}(a, E)$
4: **if** ok **then**
5: **assert** $V(c, a).\text{ok}$; **return** $\text{Commit}(c, a)$
6: **if** $j = K$ **then return** $\text{Fallback}(c)$
7: $a \leftarrow \text{Repair}(c, a, E)$; $j \leftarrow j + 1$
8: **until terminal**

Fig. 3. Validate–repair–commit procedure. At most $K+1$ candidate attempts are recorded; the successful path performs a defensive validation before mutation.

TABLE IV
PER-ERROR-CLASS EDITS OF THE GUIDED OPERATOR $\rho$

| Validator error class | Edit $\delta(e)$ on the candidate |
|---|---|
| Policy precondition omission | insert the named predicate |
| Policy effect omission | insert the named effect |
| Policy effect violation | drop the named effect |
| Unsatisfied precondition | drop the named predicate |
| Unauthorized stage mutation | clear `quest_stage_effect` |
| Unknown action type | no-op (no registered alternative) |
| Unreachable object; knowledge, disclosure, and stage classes | no-op (candidate-local edit would launder or needs state) |

TABLE V
EVIDENCE LANES AT A GLANCE: DESIGN, UNIT, COMPARISON, AND CEILING (GENERATED FROM THE EXPERIMENT MATRIX)

| Lane | Design | Unit / budget | Comparison | Ceiling |
|---|---|---|---|---|
| E1 | Offline conformance: authored frozen fixtures, one world | 13 gate; 12 repair/arm; 10 faults; 7 adapter; 3 guards | reject vs blind vs $\rho$ vs oracle | mechanism conformance |
| E2 | Screening-pilot-only RQ2: one hosted proposer, matched candidate | 5 cells $\times$ 5 calls; $K{=}1$ | $\rho$ vs blind per regime | screening-pilot-only transfer screen |
| E3 | KG/ontology simulation: closed-world typed-link holdout | 43 nodes; 106 ref. edges; 24 typed; 210 scores | degree baseline vs 6 fixed strategies | construction result only |
| ENG1 | Engineering-only Godot/Web: engine-local policy mirror, no participants | 4 fixtures; 52 checks; 8 smoke; 5 archetypes | authored fixture paths | engineering-only conformance |

Lanes are non-interchangeable with different denominators; E2 is screening-pilot-only and ENG1 is engineering-only. None establishes population efficacy, model ranking, player experience, retrieval benefit, or commercial-engine performance.

`127.0.0.1:8877`.

*1) NPC-DISC-1 disclosure contract and authority boundary:* NPC-DISC-1's 60 declaration-derived `reveal_hint` cases do not rerun/enlarge historical `verified-run-006`. Generated results/Tables X–XII define strata/bindings/input-to-check mappings; Godot rechecks service bindings before mutation.

### C. Validate–Repair–Commit

In Fig. 3's dependency-free pseudocode, Record appends in memory; persistence follows terminal consistency. $(\text{ok}, E)$ abbreviates $V$ with $\text{ok} := \bigwedge_{i=1}^{7}(v_i = 1)$; $V(c, a).\text{ok}$ revalidates.

Repair cannot mutate state; exhaustion preserves it. Typed `counterexample_guided_repair` reads only candidate $a$/errors $E$ (its sole state-derived input), never `WorldState`:

$$\rho(a, E) = a \oplus \bigcup_{e \in E} \delta(e), \tag{3}$$

Table IV's $\oplus$ applies deterministic $\delta(e)$: one field/error entity, at most $|E|$/attempt; conflicting requirements/rejections leave fields unchanged. Fixed points require identical $E$, not convergence. The nondeployable state-reading oracle ignores errors, reconstructs preconditions/effects/both quest-stage fields from state, and never launders violations by deleting required objects/used facts/disclosed facts.

### D. Integrity, Semantic Replay, and Accounting

Unkeyed SHA-256 checksums/trace links detect accidental/tested changes, not writer identity or joint content/checksum rewriting. Replay strictly parses prior state/candidate/validation, requires invalid nonfinal attempts, recomputes final state, checks equality/adjacent JSONL continuity, and never reruns repair. Request→decision→result and first-issued-action/exactly-one-receipt binding prohibit neither duplicate-action-ID issuance nor impersonation.

Commit, symbolic fallback, adapter failure, controller failure are terminal classes; controller failure preserves state/assignment without a completed proposal trace. Pre-execution keys bind run, arm, scenario, seed, model, revision, configuration/input/prior-state hashes. Aggregation rejects duplicate/missing/unexpected keys and separates failures.

## V. EVIDENCE LANES AND BOUNDARIES

Table V defines lane designs, units, comparisons, ceilings.

### A. *E1 Offline Conformance Design*

PQ1–PQ5 test immutability, frozen-class repair, checksum/semantic/linkage/continuity faults and second-append rollback, fail-closed assignment/adapter/manifest accounting, proposal/replay unknown-key rejection.

Generated results enumerate the 13 gate/12 repair fixtures and all repairability classes/arms. Latency/token constants test accounting, not population/detector performance; packets bind fixtures, assignments, schemas, results, code.

### B. *E2 Live RQ2 Screening*

E2 shares each of five seed-indexed hosted `gpt-5.6-sol` candidates per cell between blind unchanged retry/$\rho$ at $K{=}1$, varying policy visibility/zero-effect routes. Uncontrolled seeds, unpinned revision, unavailable tokens, and constructed states permit exact counts/failure classes only, not population statistics/model ranking.

### C. NPC-ENG1/NPC-SCR1: 88 Assignments, 28 Definitions

Stable v5: 16 engineering/12 screening recipes ($12 \times 2 \times 3 = 72$; combined $R/P/O = 40/24/24$). Checks: bootstrap, identity/hash, intent/plan/role/result, negative/stale/duplicate/deadline, replay. P uses authenticated Codex OAuth without fallback; O is 100% auxiliary-source, not independent gold. Status/count/latency/reproduction: `research/academic-pipeline/npc-goal-recovery/verified-run-006/`, runner/validator; verified-run-002/003/004 are history, preflight outside NPC-SCR1.

### D. *E3 KG/Ontology Link-Proposal Simulation*

E3 is a non-authoritative closed-world simulation: 43 nodes/106 reference edges/24 reviewed typed edges pass type/domain/range checks and six competency questions. Six graph-edge-withheld authored targets score seven fixed strategies and select/summarize the retained strategy, not an untouched/external test. Typed-lexical S2 recovered 6/6 ($P = R = F_1 = 1.000$, realistic-tie $\mathrm{MRR} = 0.944$, Brier $= 0.131$, $\mathrm{Sem@3} = 1.000$); degree baseline: none. These construction results establish neither OWL/SHACL conformance, semantic completeness, runtime retrieval, nor efficacy.

## VI. Pilot Results

### A. Declared-Field Gate Conformance

The pilot loader admits a fixture set only when it isolates every implemented validator code exactly once alongside one valid control, so the 13/13 agreement and the observation of all 12 implemented error codes are *construction invariants of the harness*, not measured success rates. What the run adds is the part the loader does not enforce: each of the 12 isolated negative fixtures showed observed agreement with its authored expected code rather than reaching a different one, and every noncommit path preserved the complete canonical state. This is implementation conformance to an authored structured-field oracle over a single world state, not accuracy against independent semantic labels.

TABLE VI
E1 Offline Conformance: Exact Repair-Arm Counts by Repairability Class (Designed Fixtures)

| Arm | Info source | Commits / cases | | | Mean edits[e] |
|---|---|---|---|---|---|
| | | G-rep.[f] | O-only[f] | Irrep.[f] | Fields |
| Rejection-only ($K{=}0$) | none | 0/5 | 0/1 | 0/6 | — |
| Blind unchanged retry | none | 0/5 | 0/1 | 0/6 | — |
| Validator-error-guided $\rho$ | error set $E$ | 5/5 | 0/1 | 0/6 | 1.0 |
| State-reading oracle | state+policy | 5/5 | 1/1 | 0/6 | 1.0 |

[e] Mean changed candidate fields over committed repairs (minimality proxy); failed arm executions left the state unchanged. E1 uses 12 initially invalid designed cases: rejection-only/blind 0/12, 0/12; validator-error-guided $\rho$ 5/12; state-reading oracle 6/12 commits. [f] Commits/cases. G-rep. = guided-repairable, O-only = oracle-only, Irrep. = irreparable (frozen repairability classes); these are not population efficacy or live-model quality.

TABLE VII
E1 Offline Conformance: Frozen-Pilot Checks and Terminal Outcomes (Rows Are Not Comparable)

| Check or terminal family | Exact outcome | Interpretation |
|---|---|---|
| Gate fixture agreement | 13/13 | authored-oracle agreement |
| Known-boundary acceptance | 2/2 | encoded limit, not a safety pass |
| Closed unknown-key rejection | 1/1 | proposal/replay parser parity |
| Detectable integrity-fault rejection | 10/10 | designated check operations |
| Known provenance-boundary acceptance | 1/1 | repair provenance unauthenticated |
| Adapter terminal classes | 1 commit; 1 fallback; 5 failures / 7 assigned | complete classification of 7 assignments |
| Manifest-guard rejection | 3/3 | injected faults failed closed |

Rows have different denominators and favorable directions; they must not be ranked as rates. Every value is an exact audit fact for designed fixtures.

### B. Repair-Arm Comparison on Designed Fixtures

The 12 initially invalid designed cases are partitioned into frozen repairability classes: 5 guided-repairable, 1 oracle-only, and 6 irreparable. Rejection-only and blind unchanged-candidate retry each committed 0/12. The validator-error-guided operator $\rho$ committed 5/5 guided-repairable cases and, by design, 0/1 oracle-only and 0/6 irreparable cases; every $\rho$ commit edited a mean of 1.0 candidate fields. $\rho$ reads only the prior candidate and typed validator errors $E$; those errors are its sole state-derived input, and every failed arm execution left the canonical state unchanged. The state-reading reference callback committed 5/5 guided-repairable and 1/1 oracle-only cases, with 0/6 irreparable. These exact counts separate the three mechanisms on the designed fixtures: blind retry never converts a validator error into a commit, $\rho$ converts exactly the cases whose error payload carries sufficient information, and the oracle additionally converts the case requiring state knowledge. The 5 committed guided cases initially carried `POLICY_PRECONDITION_OMISSION+UNSATISFIED_PRECONDITION`, `POLICY_EFFECT_OMISSION`, `POLICY_EFFECT_VIOLATION`,

UNSATISFIED_PRECONDITION, POLICY_QUEST_STAGE_EFFECT_VIOLATION, respectively, and each became valid through the corresponding Table IV edit $\delta(e)$ alone. This is operator conformance on frozen fixtures, not repair quality of any live model.

### C. Integrity, Boundaries, and Assignment Accounting

All 10 faults prespecified as detectable were rejected by their designated check operations (10/10). Because this pilot does not compare stable typed detector codes, it does not establish detector-layer attribution. Separately, the known provenance-boundary fixture substituted a different invalid precursor before the same recorded valid repair, recomputed the checksum, and passed replay as expected (1/1 undetected). Thus, the mechanism does not protect against a writer who can recompute unkeyed checksums, and replay does not authenticate the repair-generation operation. Two open boundary sentinels—unextracted narrative disclosure and simultaneous omission of a required object from candidate and policy—were intentionally accepted at the encoded layer (2/2), with 0 labelled as safety passes. The former unknown-top-level-field acceptance boundary was later reclassified as a closed negative regression: both proposal and replay parsers specifically rejected a complete 12-field candidate carrying one unknown key (1/1). This establishes parser rejection parity, not semantic safety. Among seven adapter/accounting assignments, outcomes were 1 commit, 1 symbolic fallback, and 5 adapter failures. Synthetic telemetry fields were populated and propagated for 2/7 assignments; these are schema-pinned constants verifying accounting-field propagation, not measurements. All 3/3 injected duplicate or missing-assignment guards failed closed. These are raw counts from synthetic offline fixtures.

### D. Live Screening Pilot (Pilot-Only)

The pilot-only counts in Table VIII show that Signal-v2/policy-blind had 5/5 invalid initial candidates; $\rho$ committed 5/5 and blind retry 0/5. The other four current cells were null, while all 15/15 noncommit outcomes preserved the prior-state hash. In the superseded v1 diagnostic, up to 3 co-occurring errors were reduced to one irreparable knowledge error but still did not commit.

TABLE VIII
E2 SCREENING-PILOT-ONLY: LIVE SCREENING EXACT COUNTS (5 CELLS, 5 PROPOSALS PER CELL, $K$=1)

| Base / condition | Initial valid | $\rho$ vs blind commits | $\Delta$ | Initial errors |
|---|---|---|---|---|
| Frozen / visible | 5/5 | 5/5 vs 5/5 | +0 | — |
| Frozen / blind | 5/5 | 5/5 vs 5/5 | +0 | — |
| Frozen / goal-blind | 0/5 | 0/5 vs 0/5 | +0 | QSR |
| Signal-v2 / visible | 5/5 | 5/5 vs 5/5 | +0 | — |
| Signal-v2 / blind | 0/5 | **5/5 vs 0/5** | **+5** | PEO+PEV+PPO |

QSR = quest-stage regression, PEO = policy-effect omission, PEV = policy-effect violation, PPO = policy-precondition omission. $\Delta$ is $\rho$ commits minus blind commits for matched candidates. There are 5 cells with 5 calls each: 3 cells were initially valid (5/5), while the goal-blind cell had QSR in 5/5 and 0/5 and 0/5 commits in both arms (null regime). Only Signal-v2/blind separated ($\rho$ 5/5 vs blind 0/5). All counts are screening-pilot-only, with no inferential statistics or model ranking.

### E. Fixed-Goal NPC Operation (NPC-ENG1/NPC-SCR1)

Mira's fixed goal is to deliver the authorized tide lead after harbor-signal restoration. Python computes bounded BFS plans from typed intent and public observations. Acquire/install remain player actions; the NPC may only commit the authorized `reveal_hint`. Godot synchronously checks session/epoch/request/goal/policy/revision/hash before calling the existing writer without an await. Results and new observations determine the next plan, unlike single-candidate validator-error repair.

NPC-ENG1 comprises 16 deterministic R-arm engineering cases initialized from actual engine snapshots (assigned_failure=2, completed=14). NPC-SCR1 retains all 72 assignments and failures, 24 per P/R/O arm (assigned_failure=12, completed=60). P records 24 authenticated Codex CLI exec attempts, 24 returned decisions, and 4 assigned failures; these are not remote billing-call counts. Requested model: unknown (null); recorded decision-model identifier: unknown (null). The latter propagates the configured selector, not a provider revision; the provider revision is unavailable. Null identifiers remain unknown; global defaults are not evidence. R uses authored rules. O is a protocol-declared auxiliary under `same_model_family_auxiliary`. Actual model-family matching is unverified. No source is independent; O is neither an oracle nor semantic gold. Table IX reports bounded protocol-conformance fixed-case counts, not recovery-efficacy, superiority, or generalization estimates. Efficacy remains pending; tokens and cost are unmeasured. A read-only CLI and isolated working directory were used, but tool-free inference and adversarial confidentiality were not established. This generation revalidates immutable verified-run-006 against archived source bytes dated 2026-10-17 (revision fa9fb8a). It is a historical packet result and does not certify the current checkout NPC runtime or current release authority. Verified pre-authored stale-control ACK rejections are P=4, R=4, O=4 per arm (SCR-T01 (stale_revision), SCR-T02 (stale_policy)); 12 instances had all corresponding negative-control gates passed. These rows remain assigned failures in the denominator, not method-performance outcomes; other assigned failures (0) remain unclassified. Replay validation is limited to the exported epoch (`intent_lineage_scope=epoch_local_only`); it does not prove intent continuity across epochs (`cross_epoch_intent_continuity_proven=false`). The opaque-anchor exception is restricted to an adjacent leading observation request/decision pair in a positive-epoch snapshot. Epoch-zero orphans, other observation requests before intent commitment, and later intent mismatches remain rejected.

### F. Frozen `reveal_hint` contract and disclosure evidence

This artifact consumes only the frozen expectation packet `sl-npc-disclosure-expectations-v1`

TABLE IX
NPC-SCR1 FIXED-CASE EXECUTION COUNTS, SEPARATE FROM THE 16 NPC-ENG1 CASES; ALL ASSIGNED FAILURES REMAIN IN THE DENOMINATOR.

| Arm | Assigned | Complete/fail | Goal reached | Recovery E/U |
|---|---|---|---|---|
| P | 24 | 20/4 | 4 | 24/0 |
| R | 24 | 20/4 | 4 | 24/0 |
| O | 24 | 20/4 | 4 | 24/0 |

E: recovery eligible; U: unreachable; NA: unmeasured. Observed goal counts are not recovery rates. Valid replay does not turn a failed assignment into a completed one. O has zero independent sources.

(v1.0.0) and four actual result files. The packet SHA-256 is `9920aa834910a3ee35ed4446de4fdba42bb67dda71b07a93bccfc7a0b2e09028`, checked directly against its sidecar. The declared normal stratum has 48 cases from `1 eligible actor x 4 declared fact IDs x 3 declared stage values x 2 knowledge values x 2 prior-disclosed values`; boundary has 4 cases and malformed inputs have 8 cases. Boundary and malformed inputs are excluded from the normal denominator, and each trace expectation is matched back to the packet by case ID rather than copied from a gate output.

The content-first explanation reads typed input (operation, actor, fact, stage, knowledge, and prior disclosure) into ontology entities/relations, applies authored rules, records actual checks, then reports an allowed/denied decision and its canonical state effect. A permitted first display adds only the requested fact (plus the declared relationship-memory entry for the tide hint), preserves stage, and increments revision once. Denials and redisplays preserve canonical state. The presentation is explanatory: it does not recover hidden facts or natural-language implications and does not claim live capture.

The runtime declaration is `sl-npc-disclosure-expectations-v1:runtime-disclosure` / v1.0.0, with provenance `frozen_expectation_packet`. Scope contains four canonical fact IDs (dock_saved, keeper_betrayal, lighthouse_dark, tide_marks_hint) and the eligible actor `captain_mira`. Rule IDs are DISC-001, DISC-002, DISC-003, DISC-004, DISC-005, DISC-006, DISC-007; the only approved templates are `mira-safe-deflection-v1` and `sl-rc-003-authorized-hint`.

The Python evaluator materializes the same cases and expectation object in three isolated arms and records semantic outcomes, check traces, errors, before/after state SHA-256 values, rendered template IDs, and replay equality. Godot is the separate game-execution boundary that owns the existing `reveal_hint` writer and canonical mutation; this producer neither reconstructs nor replaces Godot state. The mapping is traceable through operation `reveal_hint`, scenario `sealed-lighthouse-v1`, actor `captain_mira`, and bound source hashes. Execution-observed hashes match the current source. 3 source hash(es) differ from the frozen baseline (game-track/godot/scripts/game3d/game_3d.gd, game-track/godot/scripts/sealed_lighthouse_machine.gd, llm-wiki/wiki/reports/2026-10-22-ontology-signals-interview.md); the packet hash is preserved and the difference is disclosed.

TABLE X
FROZEN REVEAL_HINT CONTRACT STRUCTURE, CONTENT, RULE IDS, AND RESULT PROVENANCE.

| Field | Value |
|---|---|
| Operation | reveal_hint |
| Scenario | sealed-lighthouse-v1 |
| Normal denominator | 48 |
| Boundary | 4 |
| Malformed | 8 |
| Canonical facts | dock_saved, keeper_betrayal, lighthouse_dark, tide_marks_hint |
| Eligible actor | captain_mira |
| Ontology | sl-npc-disclosure-expectations-v1:runtime-disclosure |
| Rule IDs | DISC-001, DISC-002, DISC-003, DISC-004, DISC-005, DISC-006, DISC-007 |
| Templates | mira-safe-deflection-v1; sl-rc-003-authorized-hint |

Expectations are not derived from gate output; denominators are read from the frozen declaration.

Table XI reports the three arms over the same 60 cases with separate normal, boundary, and malformed denominators. Violations in unchecked are intentional isolated observations; general-rule and ontology-gate are separately checked for 0 frozen-expectation mismatches. Figure 4 distinguishes expected-match/replay case counts (E/R), check-record PASS/FAIL counts (C), and counts of cases with no reported checks (NR). NR does not mean skipped, pass, or out-of-scope. Table XII maps two rule-change fixtures and four paired cases to disposable policy clones; the shipped scenario is not modified.

Claims are limited to bounded engineering conformance. This evidence does not establish human gold labels, an access-separated holdout, efficacy, superiority, generalization, player benefit, model latent state, or cost. Exact input paths, hashes, and reproduction commands are bound by `paper/latex/generated/disclosure_producer_manifest.json`. Disclosure sequence/`t_ms`, public-safe hashes, categorical checks are an engine timeline, not provider receipt/latency or fixed-case counts. Missing/out-of-scope is not PASS/SKIPPED.

**ENG1.** Legacy Godot 4.7.1 replays authored rejection/commit to prior/offline hashes; no live transport. Presentation-only captures show committed board/hints/coach/chain prompt and dashboard checks/holds/unreached/out-of-scope. Core 52/52 plus four extensions establish engine-local conformance, not model/Python transport, participant, narrative, usability, efficacy.

### G. Single-task Codex development observation

Frozen `ontology-map-20261023-01`: one read-only bilingual dashboard task ($n$=1, independent repeats 0), not NPC trials/60 disclosure fixtures. Six native CLI invocations (initial + five resumes; one read-only/no-edit) are one case. Native `turn_context.model`: `gpt-6.1-sol`; provider identity/immutable revision/cost: `null`. Historical hosted `gpt-5.6-sol`/NPC denominators are unchanged; cumulative tokens are not summed.

Invocation six repaired inherited 660 px mobile-table minimum width: independent geometry RED (2 failed/35

TABLE XI
RAW-TRACE OBSERVATIONS FOR THE THREE DISCLOSURE ARMS WITH SEPARATE NORMAL, BOUNDARY, AND MALFORMED DENOMINATORS.

| Arm | $n$ (N/B/M) | Allow/Deny | Semantic mismatch | State/Display fail | Replay fail | Unique unsafe |
|---|---|---|---|---|---|---|
| unchecked_isolated | 48/4/8 | 60/0 | 60 | 0/0 | 0 | 51 |
| general_rule | 48/4/8 | 16/44 | 0 | 0/0 | 0 | 0 |
| ontology_gate | 48/4/8 | 16/44 | 0 | 0/0 | 0 | 0 |

N/B/M = normal/boundary/malformed. Semantic mismatch compares the complete expected outcome; unique unsafe cases are the union of unauthorized allows and redisplay mutations. Permitted fresh-disclosure representation mismatches alone are not unsafe. Unchecked observations do not establish efficacy, superiority, or generalization.

TABLE XII
RULE-CHANGE FIXTURE TRACEABILITY AND EDIT SCOPE FOR PAIRED CASES; THE SHIPPED SCENARIO IS NOT MODIFIED.

| Fixture | Mutation | Paired cases | Changed paths | Missed cases | Match |
|---|---|---|---|---|---|
| RC-001-stage-gate-2-to-1 | disclosure_policy.stage_gates[0].minimum_stage 2→1 | N-043, N-044 | disclosure_policy.stage_gates[0].minimum_stage | none | PASS |
| RC-002-forbid-tide-marks-hint | disclosure_policy.permanently_forbidden ['keeper_betrayal']→['keeper_betrayal', 'tide_marks_hint'] | N-047, N-048 | disclosure_policy.permanently_forbidden[1] | none | PASS |

Each fixture applies the same mutation to general_rule and ontology_gate. PASS requires no missed case and mutation_scope_preserved=true.

passed), final map/signal/Web 62/62 including cell/paragraph geometry; actual Godot Web build PASS. Continuous real-input browser attempt 10 passed; 01–09 remain (01–08 setup/route failures; 09 passed older gates with clipping). Final $(\text{seq}, t_{\text{ms}})$: stage-0 denial $(68, 19468)$ (equal hashes), fresh disclosure $(239, 58234)$ (different), authorized redisplay $(249, 59793)$ (equal). Same-session $1440\times900/390\times900$, eight roles (7 Python predicates, 6 Godot families, 7 rules), keyboard Enter, inspection immutability, frozen-comparison invariance, zero console/page/network errors/warnings from pre-navigation: **accepted under the frozen task**, not efficacy. Privacy/source-binding failures and same-thread repairs are process observations, not extra $n$. Jev: `proposal_only`, `evidence_supports=no`, $p(\text{yes}) = .27$ (`jev-1.14.0`, unvalidated), advisory, not approval/gold. No baseline, productivity, superiority, fun, human-efficacy claim. Report: `research/development-cases/ontology-map-20261023-01/result.json`; adjacent frozen `contract.json`.

## VII. DISCUSSION, THREATS, ETHICS, AND ARTIFACT SCOPE

Unexecuted plans: preregistered multi-model/oracle/blind-retry comparisons, matched-validity retrieval/memory ablations, blinded matched-validity human study with justified power/mixed-effects; screening replaces none [40], [37], [45].

**L1:** undeclared semantics/narrative quality unmeasured [58]; **L2:** one world/live state, no cross-world/policy/scale inference; **L3:** oracle/classes/regimes prove neither repair quality nor sample efficiency; **L4:** unpinned hosted revision/seeds, unavailable tokens, no E2 Python–Godot authorization round trip; **L5:** no participant, affect, comprehension, agency, enjoyment, immersion, benefit evidence; **L6:** unkeyed integrity, not authentication/tamper resistance; **L7:** unverified multiwriter consistency; **L8:** startup-dominated Godot samples $> 16.7$ ms, portability/performance untested; **L9:** selected fixtures, no population/safety-rate/model-ranking inference. Validity needs independent policy expectations/no leakage; reproducibility needs pinned artifacts/software. NPC-DISC-1's frozen IDs/templates/state axes establish no free-form/implicit, hidden-payload, or all-UI-channel leakage guarantee. Parity/counts prove neither ontology superiority/completeness/necessity, latent-state inference, cost equality, semantic gold, nor accuracy ceilings. Declaration expectations are not human gold/holdout; Jev has no label/commit authority. NPC deadline/duplicate/stale/late-drop/budget/bindings test protocol, not latency/quality. Cancellation is local-only; `core.cancel` is unit/controller-only, not human/cross-world/independent-gold/C6 evidence. No participants/personal telemetry. Future human/affect work requires ethics review, consent, compensation disclosure, minimization, retention/deletion, opt-out, calibrated/bias-controlled judges [40], [41], [42], [43].

## VIII. CONCLUSION

TRACE-RPG gates mutation/audits replay. Constructed-regime repair differs 5/5–0/5; finite-strata/three-arm NPC-DISC-1 adds disclosure/unchanged redisplay. NPC, historical `native012`, legacy ENG1 remain separate (Sec. V).

## DISCLOSURE OF AI USAGE

LLMs assisted prose/code/tests/citations; hosted-model candidates were labelled. Curated AI art (portrait, seals, textures, board, icons, guide)/desk mesh has file provenance; rendered snapshots/hashes/ledger are presentation-only. Authors checked sources/accept responsibility. NPC inputs/tables retain assignments/project-protocol-schema hashes and unknown tokens/cost/model-revision.

## DATA AND CODE AVAILABILITY

Implementation/builds, frozen packets, Godot slice, deterministic evaluation, machine-checked contribution, reference-topic, and experiment matrices: research/academic-pipeline/. paper/latex/generated/disclosure_producer_manifest.json binds

## EXTERNAL TRACE COUNTS

| | N normal | B boundary | M malformed |
|---|---|---|---|
| **U** | E 0/48<br>n=48<br>R 48/0<br>C 0/0<br>NR 48 | E 0/4<br>n=4<br>R 4/0<br>C 0/0<br>NR 4 | E 0/8<br>n=8<br>R 8/0<br>C 0/0<br>NR 8 |
| **G** | E 48/0<br>n=48<br>R 48/0<br>C 14/68<br>NR 0 | E 4/0<br>n=4<br>R 4/0<br>C 2/4<br>NR 0 | E 8/0<br>n=8<br>R 8/0<br>C 0/16<br>NR 0 |
| **O** | E 48/0<br>n=48<br>R 48/0<br>C 14/68<br>NR 0 | E 4/0<br>n=4<br>R 4/0<br>C 2/4<br>NR 0 | E 8/0<br>n=8<br>R 8/0<br>C 0/16<br>NR 0 |

U unchecked_isolated · G general_rule · O ontology_gate

E expected P/F (n cases) · R replay P/F

C records P/F · NR cases with no reported checks

P pass · F fail; NR unreported, not pass/skipped/out-of-scope.

Fig. 4. NPC-DISC-1 trace counts across three arms and normal/boundary/malformed strata. E/R count expected-match/replay cases; C counts policy-check PASS/FAIL records; NR counts cases without reported checks, not skipped checks. These are packet-bound engineering counts, not latent-state, free-language-leakage, ontology-superiority, or player-benefit measures.

research/npc-disclosure/results-signal-explainer-20261023-r5/ to frozen expectations/sidecar, bilingual inputs, fig_disclosure_signals.pdf. Earlier directories: historical receipts, not current-runtime proof; no reviewer archive/DOI deposit.

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